

SIGNATURE VERIFICATION MODEL EVALUATION REPORT

1. EXECUTIVE SUMMARY

This report details the performance of two signature verification models: the Baseline CNN (single-image classification) and the Siamese Network (pairwise comparison). Both models were evaluated on an unseen test dataset.

The Siamese Network is the recommended model. While both models have been tuned to be highly sensitive (catching ~95% of forgeries), the Siamese Network demonstrates superior overall accuracy (67.1% vs 58.8%) and a much better F1-score (75.2% vs 60.5%) at detecting forged signatures.

2. BASELINE CNN PERFORMANCE (SINGLE IMAGE)

Test Set Size: 165 signatures (110 Genuine, 55 Forged)

The Baseline model evaluates a single signature in isolation to determine if it is forged. Performance is optimized at a confidence threshold of 0.293.

Key Metrics (at optimized threshold 0.293):

- **Accuracy:** 58.79% (Overall percentage of correct predictions)
- **Precision:** 44.44% (When it flags a forgery, it is correct 44% of the time)
- **Recall:** 94.55% (Out of all actual forgeries, it successfully catches 94.5%)
- **F1-Score:** 60.47% (Harmonic mean of precision and recall)
- **ROC AUC:** 65.27% (General ability to distinguish between classes)

Confusion Matrix Breakdown:

- **True Positives** (Successfully caught forgeries): 52
- **True Negatives** (Successfully verified genuine): 45
- **False Positives** (Genuine mistakenly flagged): 65
- **False Negatives** (Forgeries that snuck through): 3

3. SIAMESE NETWORK PERFORMANCE (PAIRWISE COMPARISON)

Test Set Size: 1045 pairs (495 Genuine-Genuine, 550 Genuine-Forged)

The Siamese model compares a test signature against a known genuine reference signature. Performance is optimized at a confidence threshold of 0.345.

Key Metrics (at optimized threshold 0.345):

- **Accuracy:** 67.08% (Overall percentage of correct predictions)
- **Precision:** 62.32% (When it flags a forgery, it is correct 62% of the time)
- **Recall:** 94.73% (Out of all actual forgeries, it successfully catches 94.7%)
- **F1-Score:** 75.18% (Harmonic mean of precision and recall)
- **ROC AUC:** 68.60% (General ability to distinguish between classes)

Confusion Matrix Breakdown:

- **True Positives** (Successfully caught forged pairs): 521
- **True Negatives** (Successfully verified match pairs): 180
- **False Positives** (Matches mistakenly flagged): 315
- **False Negatives** (Forged pairs that snuck through): 29

4. CONCLUSION AND RECOMMENDATIONS

Both models exhibit extremely high Recall (~95%), meaning they are highly secure and rarely let a forged signature slip through. However, this comes at the cost of False Positives (genuine signatures being flagged for review).

The Siamese Network is significantly better at minimizing these False Positives compared to the Baseline CNN (yielding a Precision of 62.3% vs 44.4%). Moving forward, the Siamese Network should be the primary architecture used for the verification pipeline.